Essential GitHub Actions — Interview Notes

� Core Official Actions (GitHub-maintained)

1. `actions/checkout@v4` ⭐ Most Important

Purpose: Clones your repository so workflows can access code
Why needed: Runners start with empty workspace [1] [2]
Basic usage:

```
- uses: actions/checkout@v4
```

Common inputs:

```
- uses: actions/checkout@v4
with:
  ref: main                       # Branch/tag/SHA to checkout
  fetch-depth: 0                  # Full git history (default: 1)
  token: ${{ secrets.PAT }}       # Custom PAT for private repos
  ssh-key: ${{ secrets.SSH_KEY }}   # SSH-based fetch
  path: ./subdir                  # Clone to subdirectory
  repository: owner/repo          # Clone different repo
```

Notes:

- Always first step in most workflows [1]
- Persists auth token in git config [3]
- Default: clones current repo, current branch, shallow clone (1 commit) [4] [3]

2. `actions/upload-artifact@v4`

Purpose: Upload files/artifacts from workflow (e.g., build outputs, test reports)
Usage:

```
- uses: actions/upload-artifact@v4
with:
  name: my-artifact
  path: dist/
  retention-days: 5
```

Notes:

- Artifacts persist across jobs in same workflow run
- Can download artifacts from workflow run UI
- Essential for passing build outputs between jobs

3. `actions/download-artifact@v4`

Purpose: Download artifacts uploaded by other jobs/runs
Usage:

```
- uses: actions/download-artifact@v4
with:
  name: my-artifact
  path: ./downloads/
```

4. `actions/cache@v4` ⭐ Important for Performance

Purpose: Cache dependencies/build outputs to speed up workflows
Usage:

```
- uses: actions/cache@v4
with:
  path: ~/.npm
  key: ${{ runner.os }}-node-${{ hashFiles('**/package-lock.json') }}
  restore-keys: |
    ${{ runner.os }}-node-
```

Notes:

- Reduces workflow time significantly
- Cache key typically includes OS + dependency hash
- Common paths: `~/.npm`, `~/.cache`, `node_modules/`, `~/.m2/repository`

5. `actions/setup-node@v4`

Purpose: Set up Node.js environment
Usage:

```
- uses: actions/setup-node@v4
with:
  node-version: '20'
  cache: 'npm'  # or 'yarn', 'pnpm'
```

Notes:

- Installs Node.js and sets up PATH
- Automatically caches `node_modules` when cache is specified

6. `actions/setup-python@v5`

Purpose: Set up Python environment
Usage:

```
- uses: actions/setup-python@v5
with:
  python-version: '3.11'
  cache: 'pip'
```

7. `actions/setup-java@v4`

Purpose: Set up Java (JDK) environment
Usage:

```
- uses: actions/setup-java@v4
with:
  java-version: '17'
  distribution: 'temurin'  # or 'adopt', 'microsoft'
  cache: 'maven'  # or 'gradle'
```

8. `actions/setup-go@v5`

Purpose: Set up Go environment
Usage:

```
- uses: actions/setup-go@v5
with:
  go-version: '1.21'
```

9. `actions/upload-release-asset@v1`

Purpose: Upload release assets (binaries) to GitHub releases
Usage:

```
- uses: actions/upload-release-asset@v1
env:
  GITHUB_TOKEN: ${{ secrets.GITHUB_TOKEN }}
with:
  upload_url: ${{ github.event.release.upload_url }}
  asset_path: ./my-app
```

```
  asset_name: my-app
  asset_content_type: application/octet-stream
```

10. `actions/stale@v9`

Purpose: Mark/close stale issues and PRs
Usage:

```
- uses: actions/stale@v9
with:
  days-before-stale: 30
  days-before-close: 7
  stale-issue-message: 'This issue is stale'
```

🐳 Container & Docker Actions

11. `docker/build-push-action@v5` ⭐ Important

Purpose: Build and push Docker images
Usage:

```
- uses: docker/build-push-action@v5
with:
  context: .
  push: true
  tags: user/app:latest
  cache-from: type=gha
  cache-to: type=gha,mode=max
```

Notes:

- Built by Docker team
- Supports BuildKit caching via GitHub Actions cache

12. `docker/login-action@v3`

Purpose: Login to Docker registry (Docker Hub, ECR, GCR, etc.)
Usage:

```
- uses: docker/login-action@v3
with:
  registry: ghcr.io
  username: ${{ github.actor }}
  password: ${{ secrets.GITHUB_TOKEN }}
```

☁ Cloud Provider Actions

13. `aws-actions/configure-aws-credentials@v4` ⭐ Important for AWS

Purpose: Configure AWS credentials for workflow
Usage:

```
- uses: aws-actions/configure-aws-credentials@v4
with:
  aws-access-key-id: ${{ secrets.AWS_ACCESS_KEY_ID }}
  aws-secret-access-key: ${{ secrets.AWS_SECRET_ACCESS_KEY }}
  aws-region: us-east-1
```

Notes:

- Sets up `AWS_ACCESS_KEY_ID`, `AWS_SECRET_ACCESS_KEY`, `AWS_DEFAULT_REGION`
- Can also use OIDC for passwordless auth (recommended)

14. `aws-actions/amazon-ecs-deploy-task-definition@v1`

Purpose: Deploy to Amazon ECS
Usage:

```
- uses: aws-actions/amazon-ecs-deploy-task-definition@v1
with:
  task-definition: task-definition.json
  service: my-service
  cluster: my-cluster
```

15. `google-github-actions/setup-gcloud@v2`

Purpose: Set up Google Cloud SDK
Usage:

```
- uses: google-github-actions/setup-gcloud@v2
with:
  service_account_key: ${{ secrets.GCP_SA_KEY }}
  project_id: ${{ secrets.GCP_PROJECT_ID }}
```

16. `azure/login@v2`

Purpose: Login to Azure
Usage:

```
- uses: azure/login@v2
with:
  creds: ${{ secrets.AZURE_CREDENTIALS }}
```

🔧 Utility & Testing Actions

17. `softprops/action-gh-release@v1` ⭐ Popular

Purpose: Create GitHub releases with assets
Usage:

```
- uses: softprops/action-gh-release@v1
with:
  files: dist/*
env:
  GITHUB_TOKEN: ${{ secrets.GITHUB_TOKEN }}
```

18. `actions/github-script@v7`

Purpose: Run GitHub API calls using Node.js/Python
Usage:

```
- uses: actions/github-script@v7
with:
  script: |
    github.rest.issues.createComment({
      issue_number: context.issue.number,
      owner: context.repo.owner,
      repo: context.repo.repo,
      body: '✅ CI passed!'
    })
```

19. `peaceiris/actions-gh-pages@v4`

Purpose: Deploy static site to GitHub Pages
Usage:

```
- uses: peaceiris/actions-gh-pages@v4
with:
  github_token: ${{ secrets.GITHUB_TOKEN }}
  publish_dir: ./docs
```

20. `codecov/codecov-action@v4`

Purpose: Upload code coverage to Codecov
Usage:

```
- uses: codecov/codecov-action@v4
with:
  file: ./coverage/lcov.info
  flags: unittests
```

21. `actions/cache/save@v4`

Purpose: Manually save cache (for advanced scenarios)

� Quick Reference Table

Action	Purpose	Version
`actions/checkout`	Clone repo	`@v4`
`actions/cache`	Cache dependencies	`@v4`
`actions/upload-artifact`	Upload build outputs	`@v4`
`actions/download-artifact`	Download artifacts	`@v4`
`actions/setup-node`	Set up Node.js	`@v4`
`actions/setup-python`	Set up Python	`@v5`
`actions/setup-java`	Set up Java	`@v4`
`actions/setup-go`	Set up Go	`@v5`
`docker/build-push-action`	Build/push Docker	`@v5`
`docker/login-action`	Docker login	`@v3`
`aws-actions/configure-aws-credentials`	AWS auth	`@v4`
`softprops/action-gh-release`	Create releases	`@v1`
`actions/stale`	Close stale issues	`@v9`
`codecov/codecov-action`	Upload coverage	`@v4`

� Key Interview Points

1. `actions/checkout@v4` is almost always the first step — runners start with empty workspace [2] [1]
2. **Version pinning:** Use `@v4` (major version) not specific commits for auto-updates within major version

3. `actions/cache` dramatically reduces CI time for dependencies
4. **Artifacts** vs **Cache**: Artifacts persist for workflow runs, cache is for speeding up repeated runs
5. `setup-*` **actions** automatically configure PATH and can cache package managers
6. `GITHUB_TOKEN` is automatically provided; use secrets for external services
7. **OIDC** is preferred over long-lived credentials for cloud providers (AWS/Azure/GCP)

⁂

1. https://notes.kodekloud.com/docs/GitHub-Actions-Certification/GitHub-Actions-Core-Concepts/Configure-Checkout-Action
2. https://dev.to/spacelift/using-checkout-action-in-github-actions-workflow-2bna
3. https://stackoverflow.com/questions/67131269/what-is-the-use-of-actions-checkout-in-github-actions
4. https://graphite.com/guides/github-actions-checkout
5. https://spacelift.io/blog/github-actions-checkout
6. https://github.com/marketplace/actions/checkout-action
7. https://github.com/actions/checkout
8. https://github.com/actions/checkout/tree/v4
9. https://github.com/marketplace/actions/checkout
10. https://github.com/actions/checkout/releases